

Digital Twin of an Ultrasonic Fermentation System

A Proof of Concept

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Context: Beer Fermentation & Ultrasound

- Beer fermentation is a **slow, resource-intensive** industrial process: experiments take days and are costly to repeat
- **Ultrasound technology** (20–50 kHz, low power) is a promising tool to accelerate yeast activity

Yeastime estimates: a 30% reduction in fermentation time in craft breweries could lead to cost savings of approximately **10–15%**

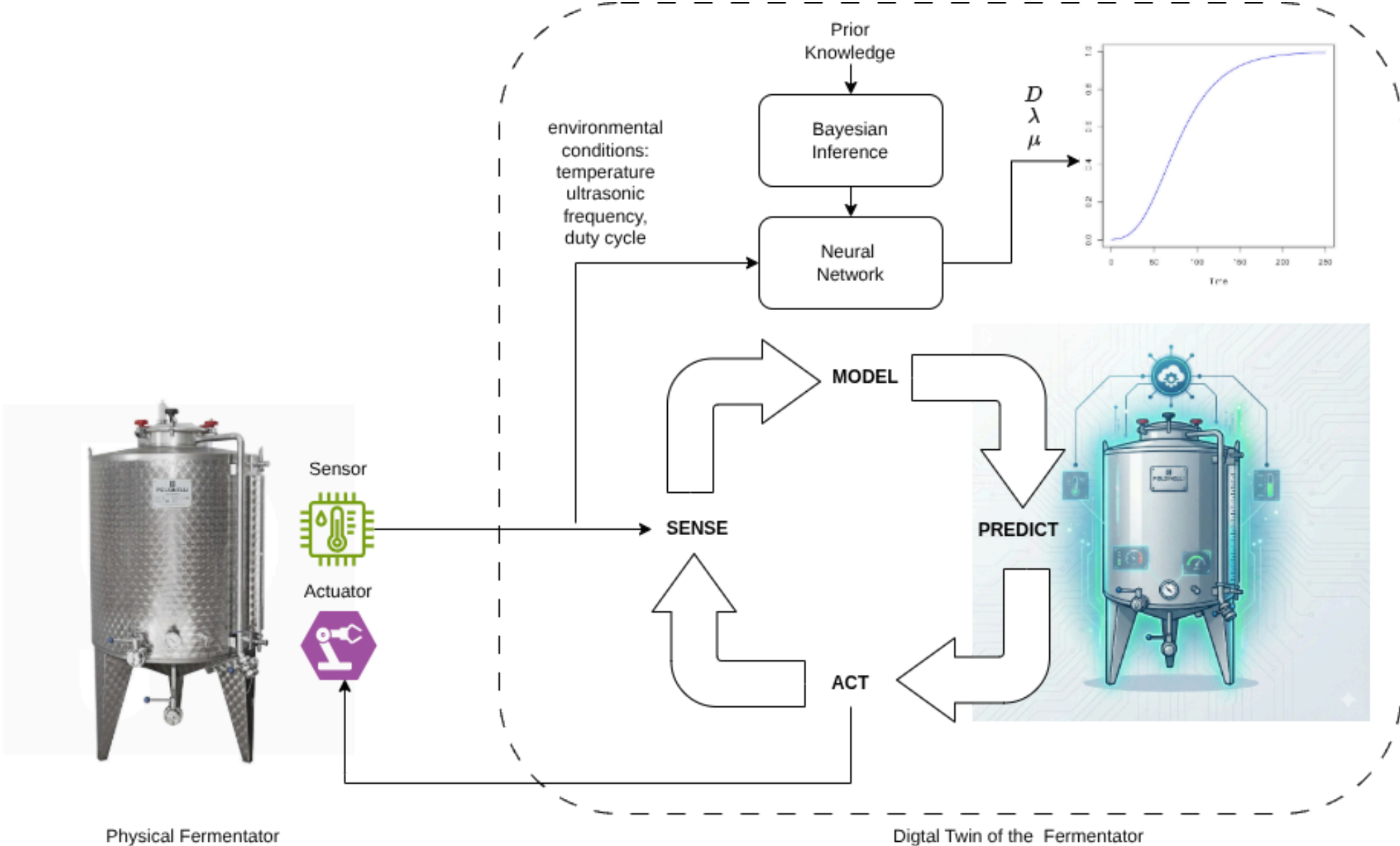
- Key effects: improved mass transfer, increased cell permeability, reduced inhibitory metabolites
- This work presents the **first digital twin tailored to an ultrasound-treated fermentation system**

What is a Digital Twin?

A **digital twin** is a virtual representation of a physical system that stays continuously synchronised through real-time data.

💡 Key Benefit

Physical fermentation trials are **slow and expensive**, the digital twin complements them with **rapid simulation (growth model) and intelligent actuation**.



The Dataset

Biological datasets are limited by **slow, manual sampling** each growth curve takes days

⚠ Data Scarcity Challenge

~100 data points: approximately **100× smaller** than the reference study [Palacios et al., 2014]

Classical training is ineffective: high risk of overfitting.

The Dataset

Feature	Description
Ultrasound duty cycle	Fraction of time transducer is active
Irradiation frequency	Ultrasonic frequency applied (normalised)
Temperature	Medium temperature (normalised)
Initial / final Optical Density	Yeast density measured at 600 nm

Points sharing the same duty cycle, frequency, and initial density belong to the **same biological growth curve**: treated as a group in cross-validation.

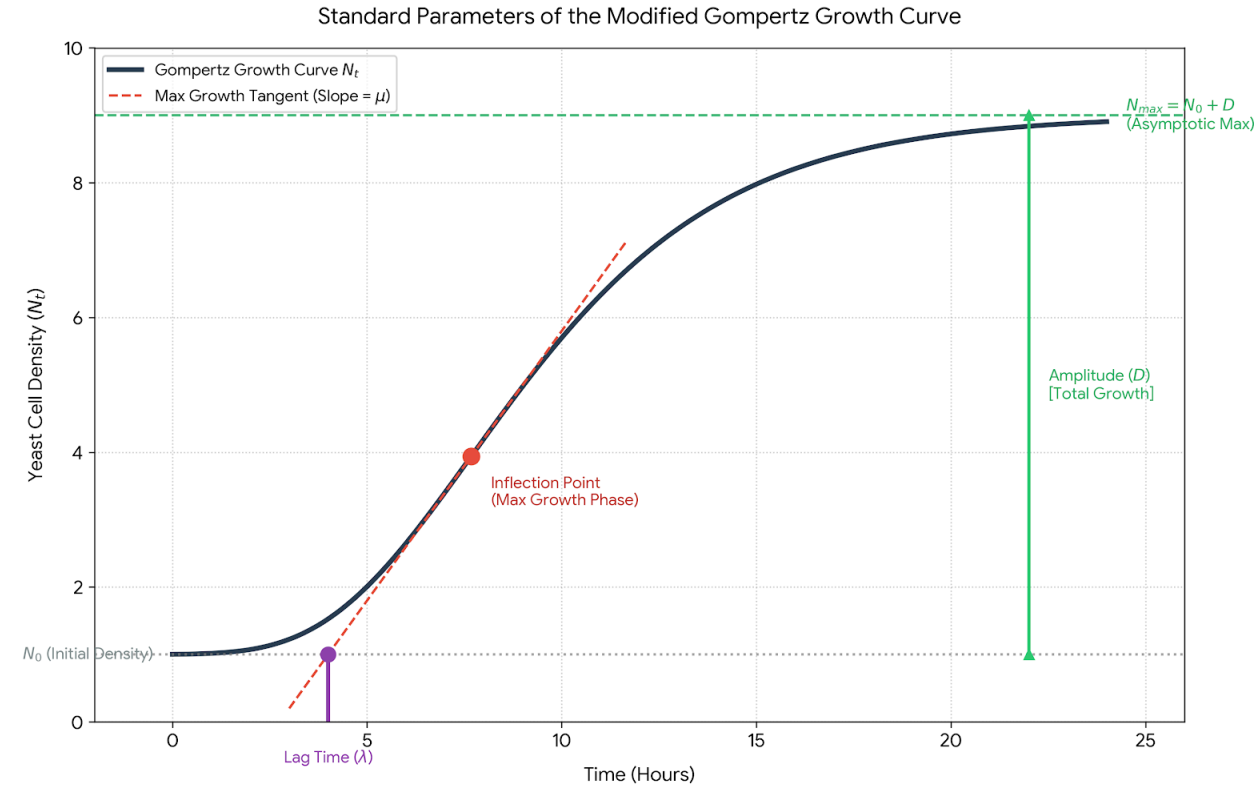
Gompertz Growth Model

- **D** = total growth amplitude
- **μ** = max growth rate
- **λ** = lag time

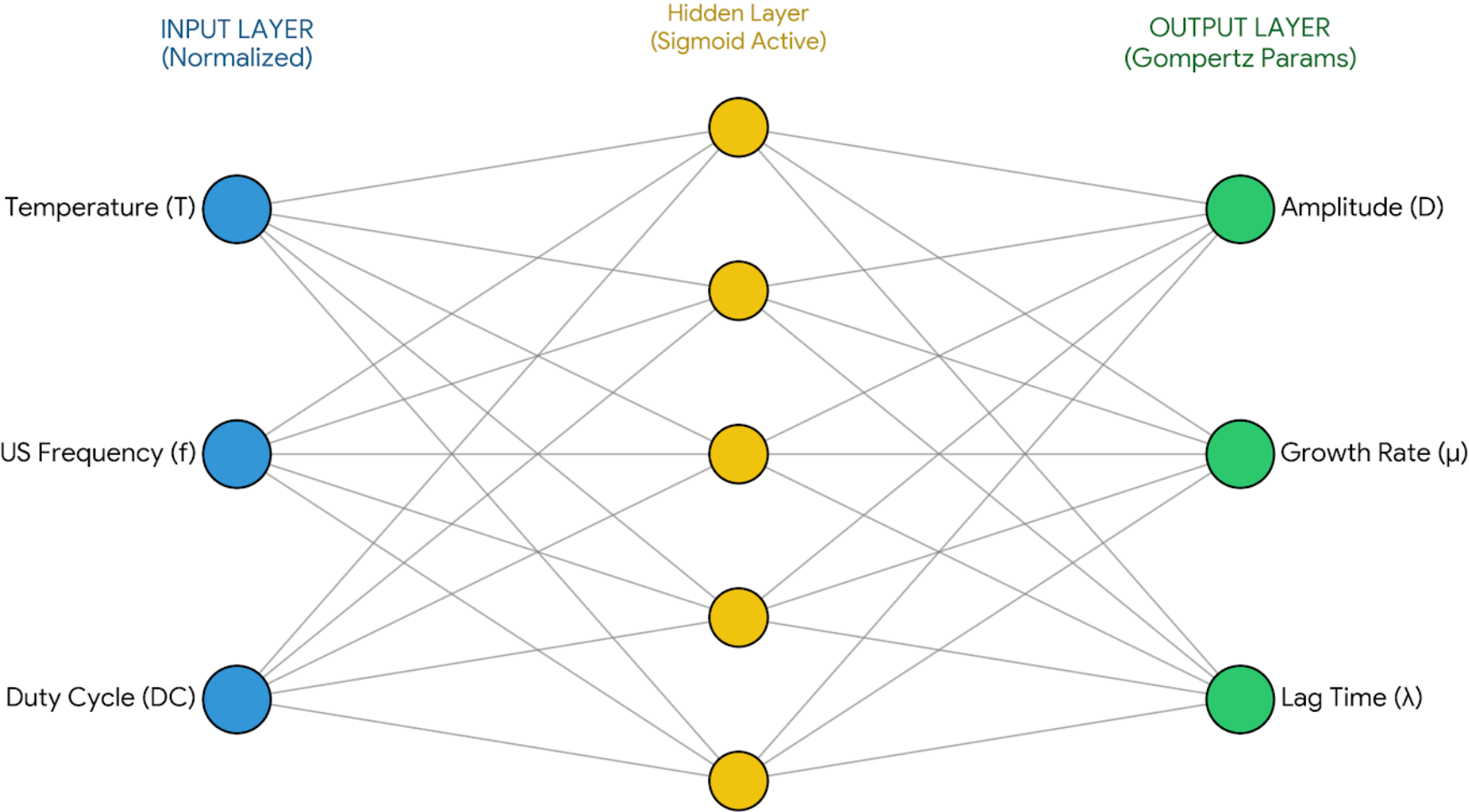
Predicted by a Bayesian Neural Network.

Yeast Density at time t:

$$E[N_t | N_0, D, \mu, \lambda] = N_0 + D \cdot \exp\left(-\exp\left(1 + \frac{\mu e(\lambda - t)}{D}\right)\right)$$



Goal



Priors

We have "clear", physical intuition for the outputs (D, μ, λ), but we have zero biological intuition for what individual neural network weights (w) should look like.

Translate a Prior on the Output Space (D, μ, λ) into a Prior on the Parameter Space (w) via the Likelihood function and MCMC sampling.

Detour

The posterior probability of a set of weights $\mathbf{w}$ given the observed fermentation data y is:

$$p(\mathbf{w} \mid y) \propto p(y \mid \mathbf{w}) \cdot p(\mathbf{w})$$

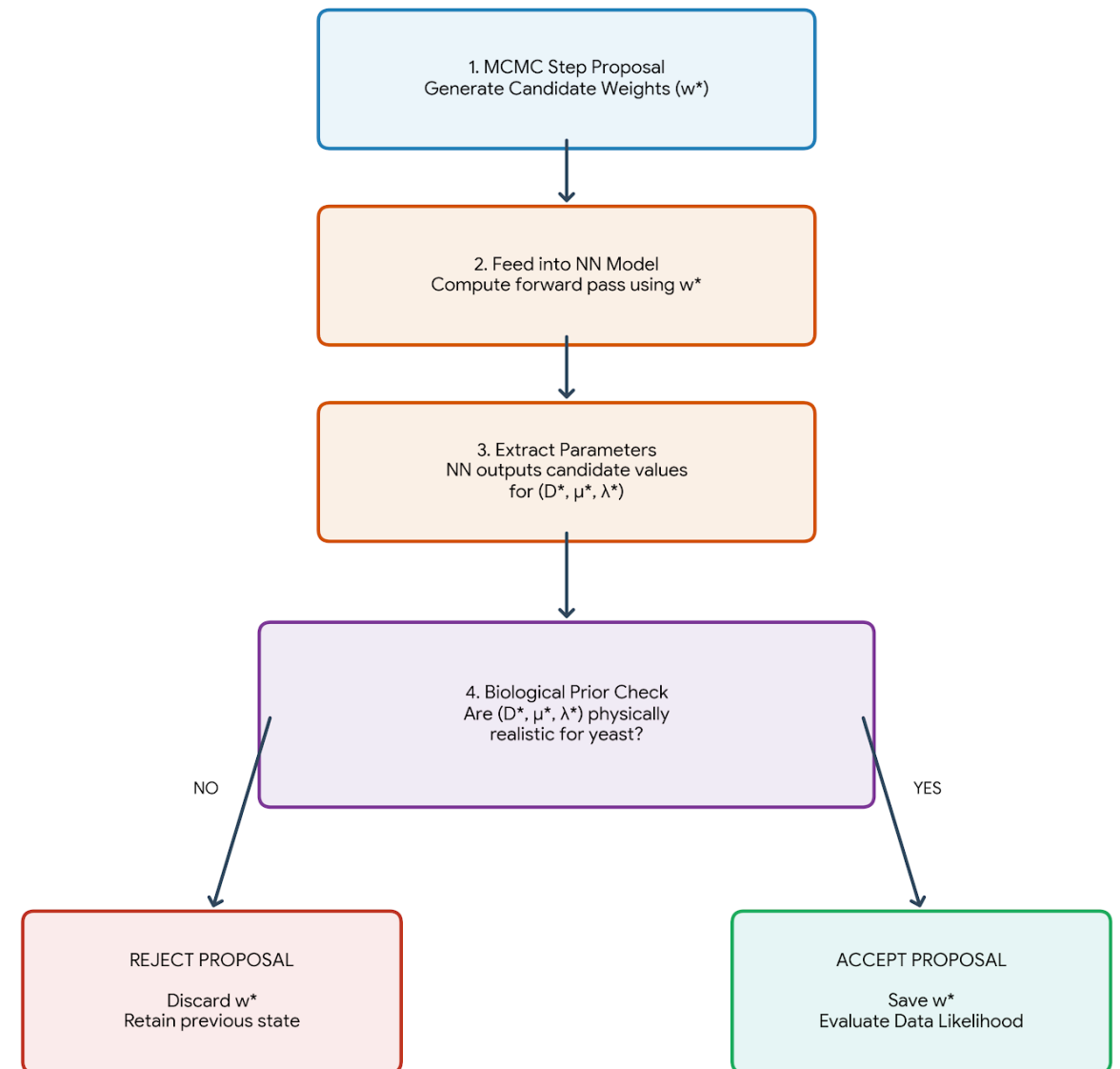
- $p(y \mid \mathbf{w})$ is the Likelihood: How well do the weights explain the raw sensor data?
- $p(\mathbf{w})$ is the Weight Prior

If a configuration of weights $\mathbf{w}^*$ causes the impossible value for D , μ , or $\lambda \implies P(\mathbf{w}^*) = 0$

The biological output limits act as a filter that shapes the allowed weight space.

Monte Carlo (MCMC) Sampling

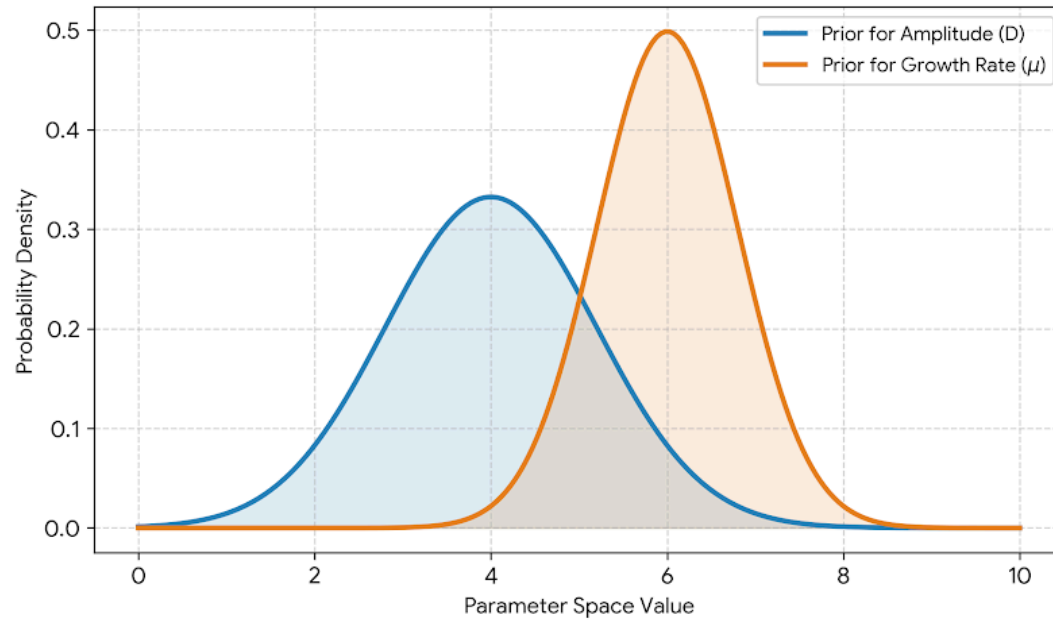
Because we can't analytically calculate how to constrain the weights to get the right outputs, we let the Random Walk Metropolis algorithm discover the valid weight spaces through exploration.



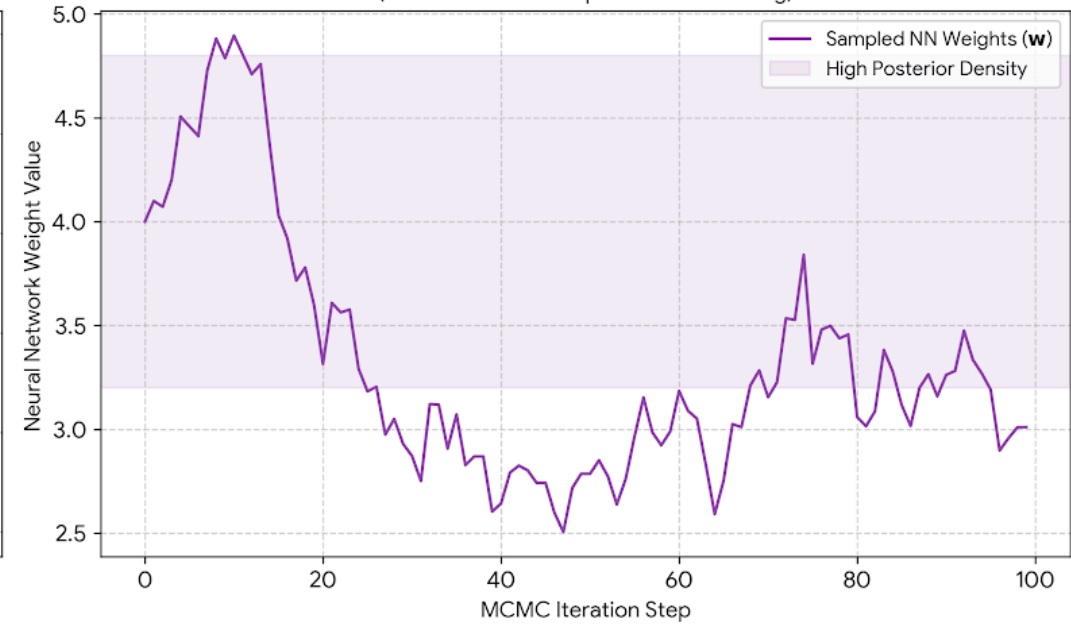
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Complete Bayesian Pipeline: From Priors and Monte Carlo Sampling to Growth Prediction

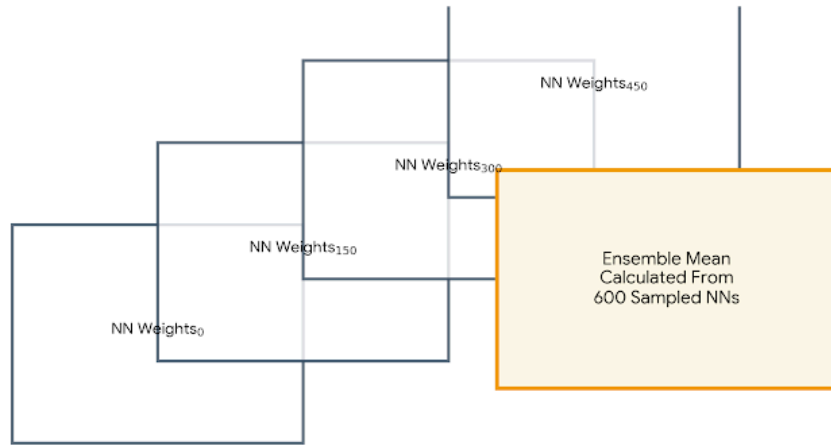
1. Informative Biological Priors
(Defining Allowed Parameter Ranges)



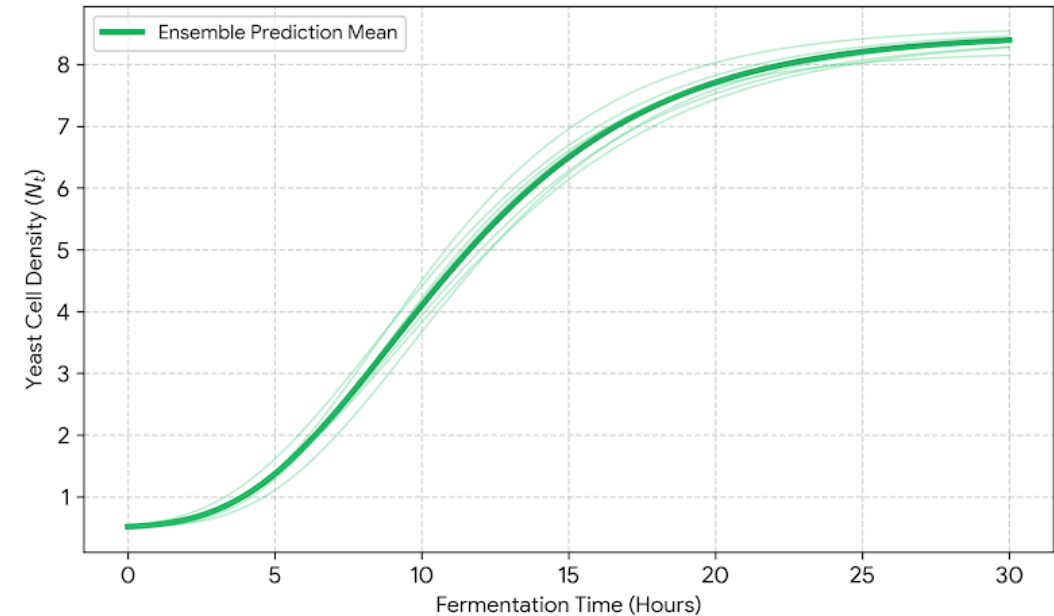
2. Monte Carlo Weight Optimization
(Random Walk Metropolis MCMC Tracing)



3. Evaluated Bayesian Ensemble Model



4. Resulting Population Growth Curves
(Gompertz Density Prediction Matrix)



Bayesian Inference Pipeline

Given the **limited dataset**, classical training is ineffective. We adopt [Palacios et al., 2014]:

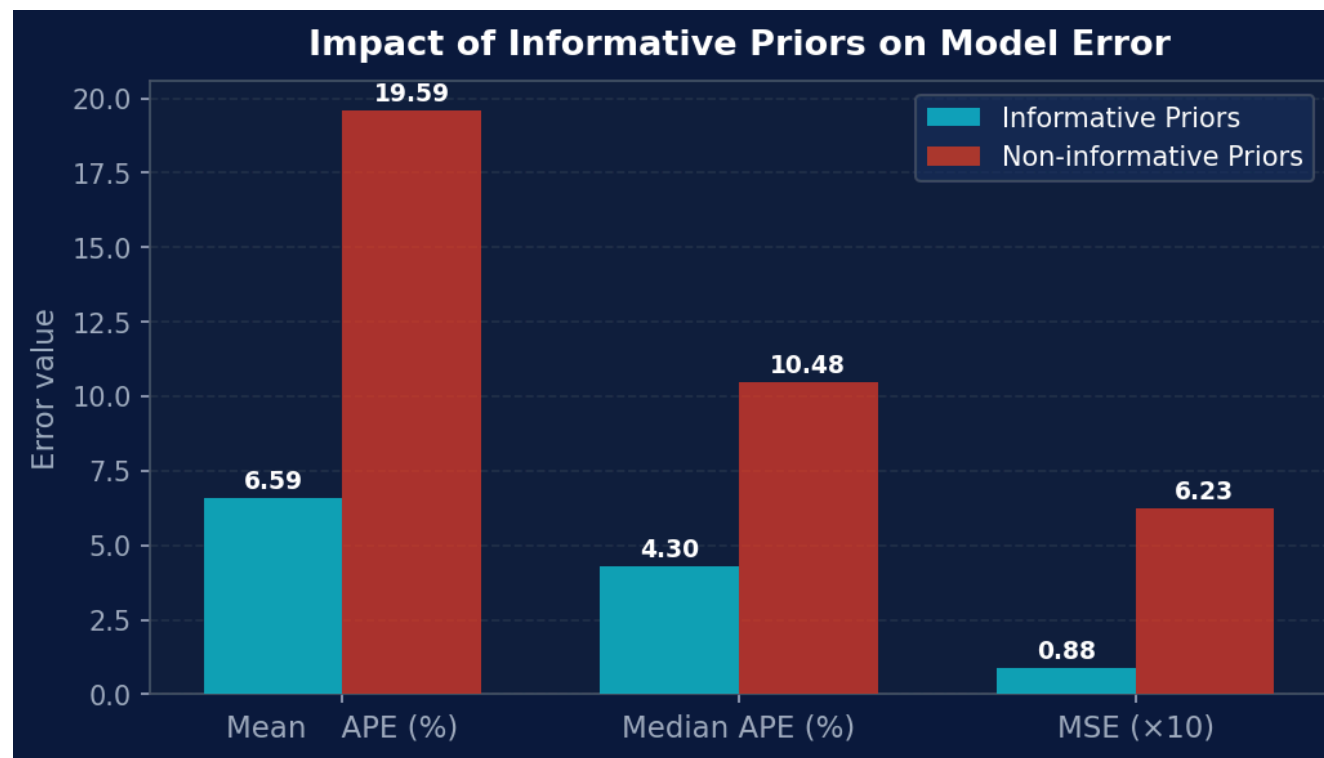
1. **Prior assignment:** probability distributions encode biological domain knowledge
2. **Random Walk Metropolis:** explores the posterior to find valid weight configurations; Metropolis-Hastings criterion escapes local maxima
3. **Network ensemble:** each accepted MCMC state instantiates a distinct NN outputting D, μ, λ
4. **Final prediction:** parameters averaged across all sampled networks → Gompertz curve

For **small datasets**, highly informative priors are essential. For larger datasets, weakly informative priors such as $\mathcal{N}(0, 100)$ are sufficient (as also recommended in [Palacios et al.]).

Informative Priors Matter

Removing informative priors causes a **~7× increase in MSE**:

- Mean Absolute Percentage Error (Mean APE)
- Median Absolute Percentage Error (Median APE)
- Mean Square Error (MSE)



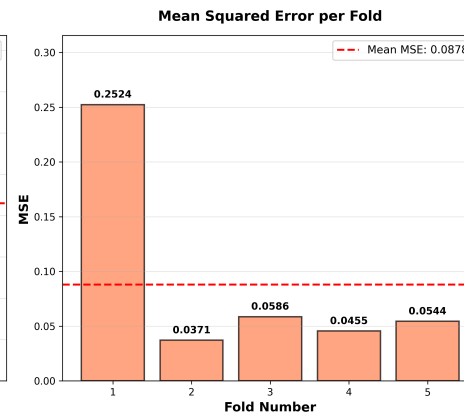
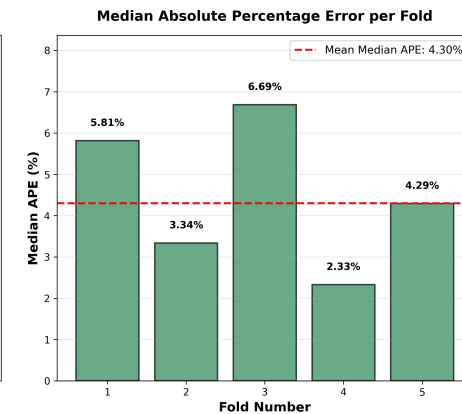
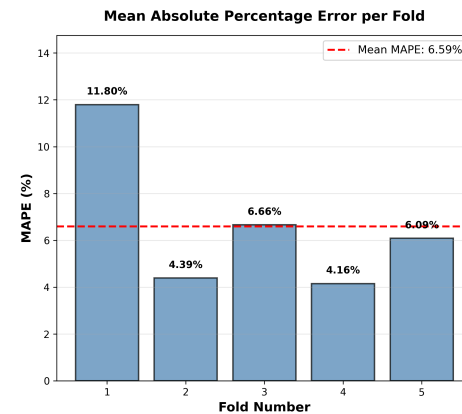
Takeaway

In fermentation, experiments are too slow to generate large datasets. **Biological prior knowledge is the only practical regulariser.**

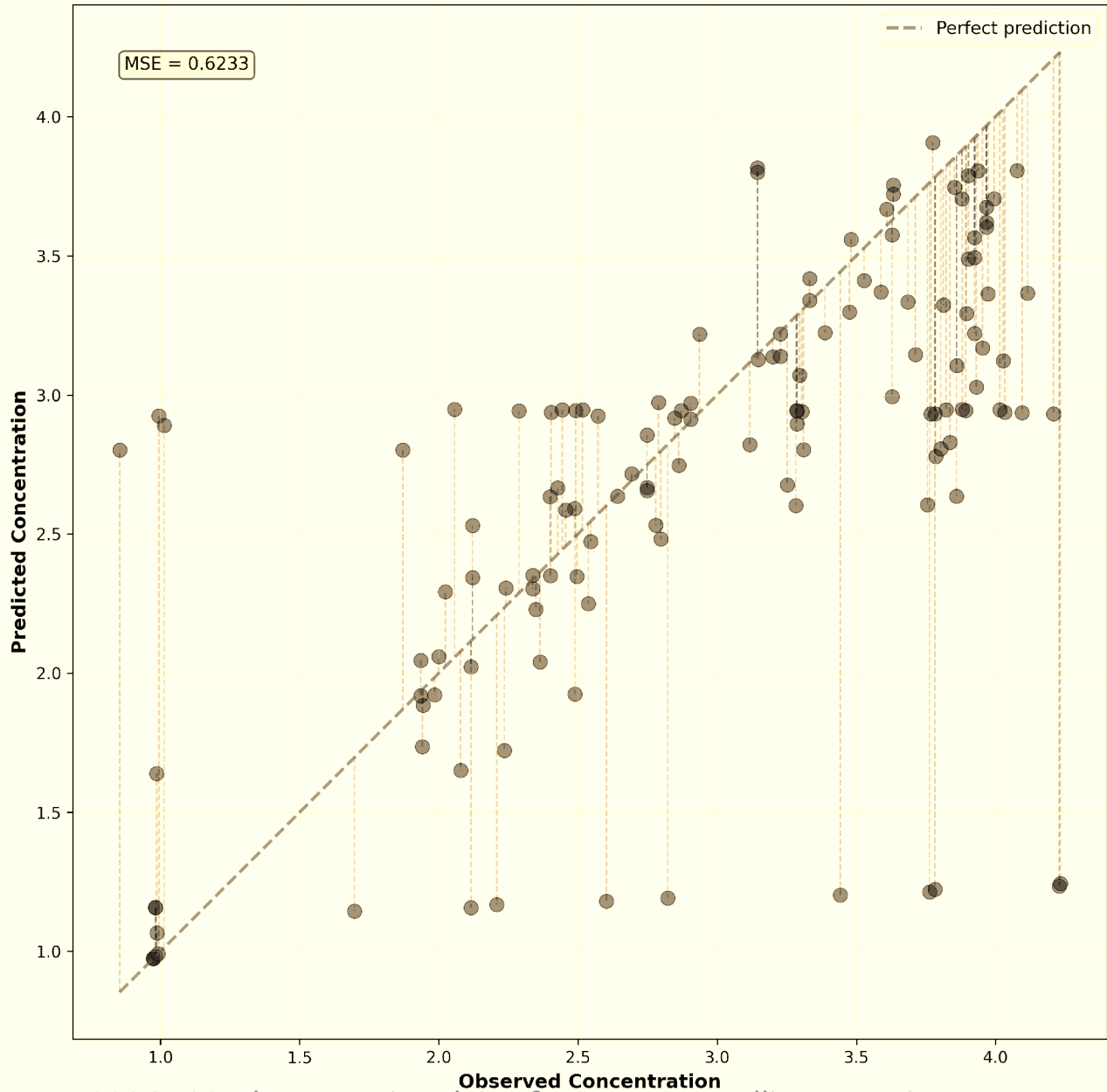
Model Performance: 5-Fold Cross-Validation

MSE is ~88× higher than [Palacios et al.]

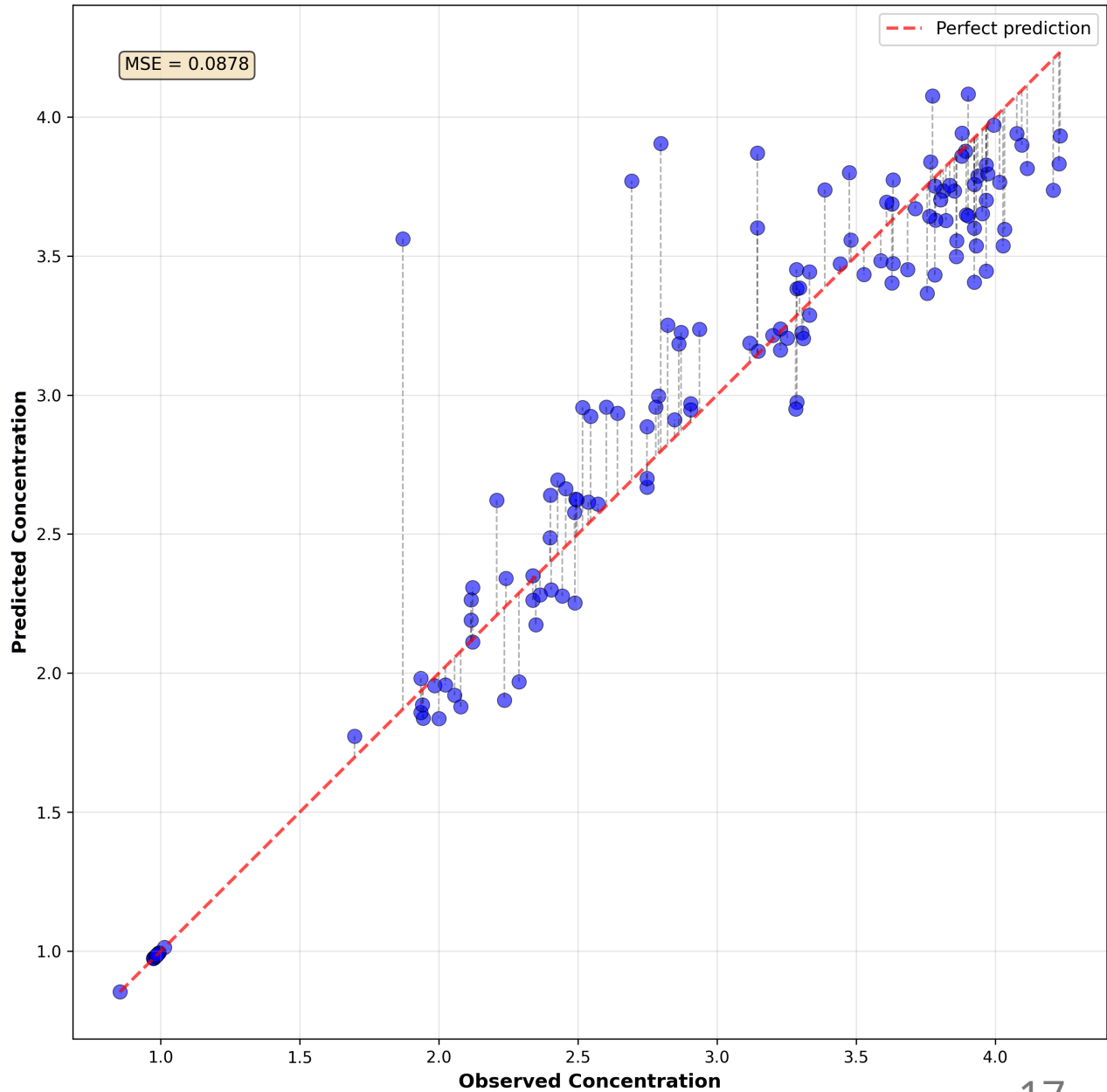
- dataset size (~100× smaller)
- single MCMC chain
- optical density (OD) measurement scale differences.

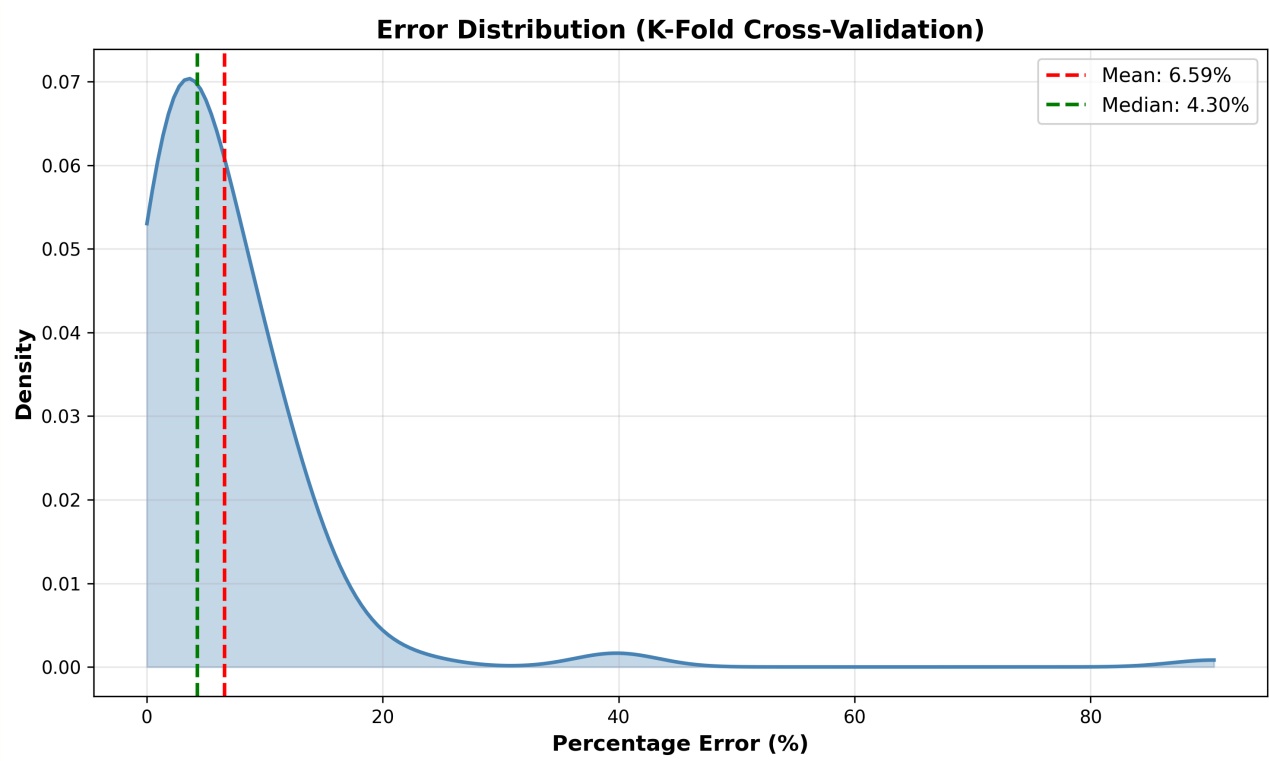
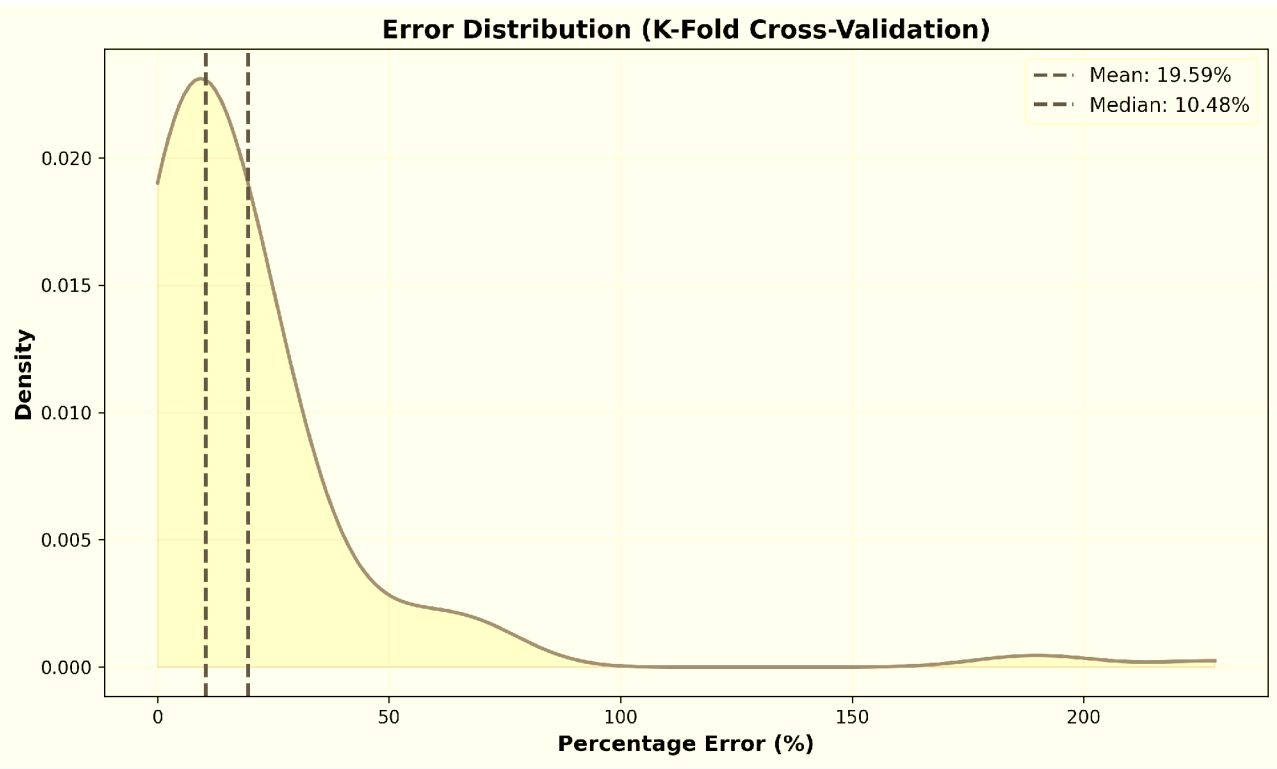


Observed vs Predicted Values (K-Fold Cross-Validation)



Observed vs Predicted Values (K-Fold Cross-Validation)

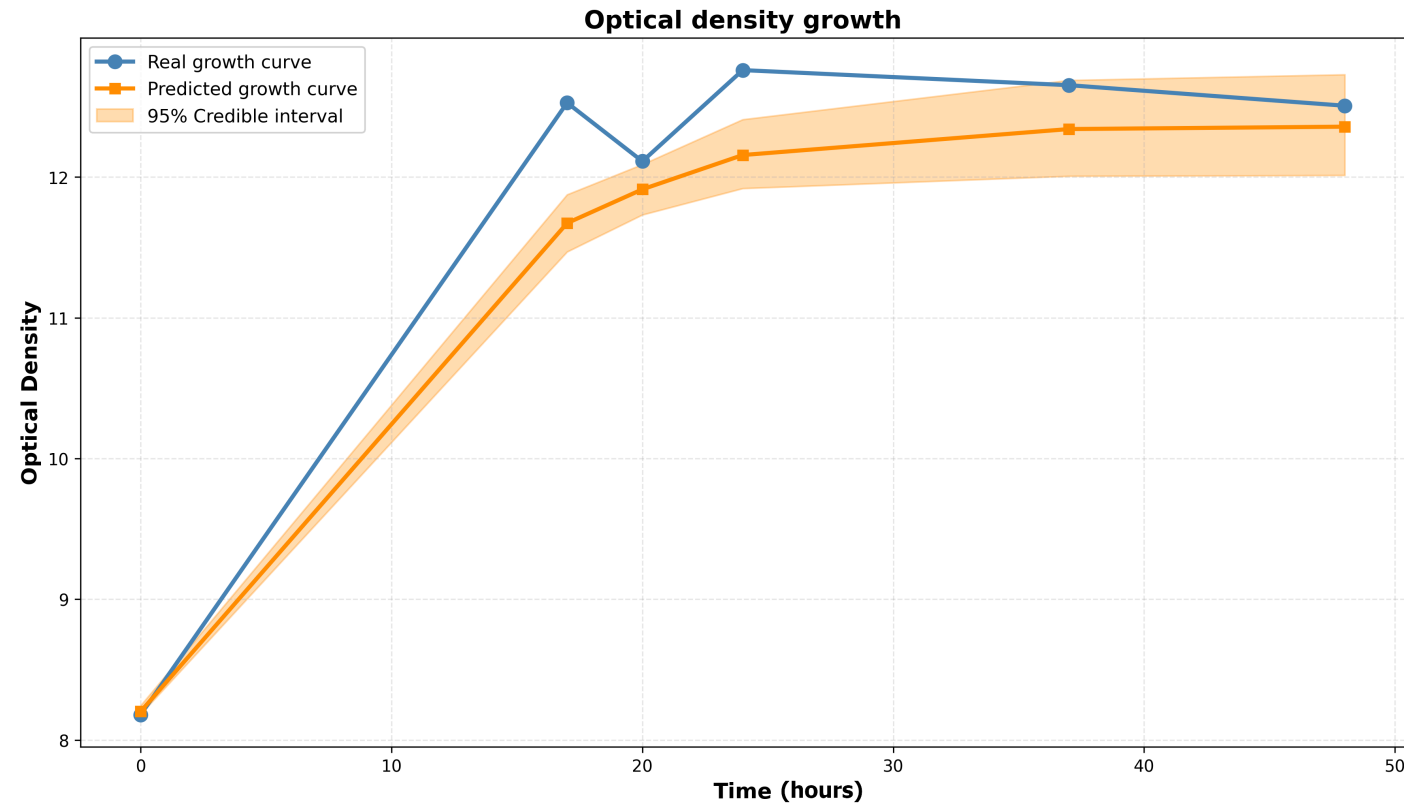




Predicted vs. Observed Growth Curve

✓ Validation

The model demonstrates Feasibility for small-data bioprocess prediction: a solid foundation for a fully autonomous digital twin.



Actuation: Closing the Loop

The digital twin controls the **ultrasonic stimulation parameters**:

-  **Frequency** — piezoelectric transducer driven at resonance via PWM square wave (ESP32 + H-bridge amplifier)
-  **Duty cycle** — delivered as energy bursts modulated at 150 Hz
-  **Temperature** — monitored; closed-loop thermal control is left for future work

Accelerating yeast growth shortens the production cycle → significant economic benefit. The digital twin enables testing actuation strategies **without costly physical trials**.

Conclusions

- First **proof of concept** digital twin for an ultrasonic fermentation system
- Bayesian NN + Gompertz model **effective under data scarcity** (~100 points)
- Informative priors reduce MSE by **~7×** vs. non-informative baseline
- Real-time sensing pipeline (ESP32, piezoelectric disk, Dallas probe) **validated on physical fermenter**

Core Message

The digital twin enables **faster, cheaper, smarter** fermentation optimisation — complementing slow physical experiments with rapid simulation and autonomous actuation.

Future Work

-  **Expand dataset.** Collect more growth curves to relax prior constraints and improve generalisation
-  **Beyond Gompertz.** Investigate alternative growth models to better capture yeast dynamics
-  **Automate OD sensing.** Currently manual; real-time density measurement is the key bottleneck for full automation
-  **Multiple MCMC chains.** Improve posterior exploration beyond the current single-chain implementation
-  **Full closed-loop validation.** Experimentally validate the feedback-optimisation framework end-to-end

**Most importantly,
enjoy your beer.**



Discussion

- How can **digital twin frameworks** best handle the tension between data scarcity and model accuracy in biological systems?
- What are the practical thresholds at which **Bayesian priors** can be relaxed in favour of data-driven learning?
- How should **ultrasonic actuation parameters** be explored efficiently — can the digital twin guide active learning?
- What does **intelligent environment** design look like for bio-oriented systems such as breweries?